AI Generative Models in Synthetic Biology:

In synthetic biology, generative AI models are increasingly used to design DNA, RNA, and protein sequences or to simulate gene circuits. The following table summarizes four major types of generative models, their applications, advantages, and limitations in the context of synthetic biology.

Model Type	Concept	Applications in Synthetic Biology	Advantages	Limitations
Variational Autoencoder (VAE)	Learns to compress sequences into a latent space and reconstruct them	Generate new DNA/protein sequences with similar functionality; design synthetic promoters	Captures hidden relationships; works well on small/noisy datasets	May generate unrealistic sequences if latent space is poorly trained
Generative Adversarial Network (GAN)	Two networks (generator & discriminator) compete to produce realistic outputs	Design synthetic sequences, optimize promoters, predict cell responses	Can learn highly realistic biological distributions	Training can be unstable; requires careful tuning
Diffusion Model	Starts from noise and gradually “denoises” to generate realistic data	Protein/molecule generation; design 3D structures; optimize stability	Produces high-quality, diverse outputs; models complex interactions	Computationally expensive; relatively new in bio applications
Transformer / GPT-based Models	Attention-based model learns patterns in sequences like language	Predict sequence function; generate DNA/protein sequences; model long-range dependencies	Excellent for long sequences and context understanding; scalable with large datasets	Requires large datasets; may overfit if data is limited